

EXCELERATE

RIT Student Participation Analysis Report

Exploratory Analysis and Predictive Modelling

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Stakeholder Report

Executive Summary

This report summarizes the analysis of the RIT student participation dataset. The work covered data cleaning, exploratory analysis and predictive modelling. The cleaned dataset contained 8,552 records and 23 columns. The analysis found that internships attracted the most participants but also had the highest number and rate of unsuccessful outcomes. Courses showed the strongest mix of in-progress and successful participation.

Three classification models were tested to predict whether a learner would be In Progress, Successful or Unsuccessful. Logistic Regression was selected as the most suitable model because it gave the best overall balance across the three groups. It achieved 87.43% accuracy and the highest macro-average F1-score of 64%. The model can help stakeholders identify learners who may require early support. Its predictions should help stakeholders make informed decisions, while final decisions should remain subject to further review.

Project Purpose

The purpose of this project was to understand student participation patterns and explore whether learner outcomes could be predicted from the available data. The analysis focused on four practical questions:

- Which opportunity categories attract the most participation?
- Where are unsuccessful outcomes most common?
- What learner and participation patterns appear in the data?
- Can a classification model help identify learners who may require support?

Data Preparation

The original dataset contained 8,558 rows. After cleaning, the final dataset contained 8,552 rows and 23 columns. Duplicate records were removed where applicable, missing values in key text fields were handled, date columns were converted into suitable formats, and inconsistent institution names were reviewed. Detailed status descriptions were also grouped into three clearer categories: In Progress, Successful and Unsuccessful.

Missing or invalid date values were kept as missing rather than being replaced with estimated dates. For predictive modelling, missing Application Year and Signup Year values were filled with their median values so the models could be trained.

Main Findings

Finding	Result
Opportunity demand	Internships were the largest category with 5,415 participants.
Internship outcomes	4,235 internship participants were unsuccessful, about 78.2% of the category.

Finding	Result
Course outcomes	Courses had the most in-progress learners and the most successful learners.
Successful outcomes	Only 29 records were Successful, and they appeared only in courses and events.
Participation year	Among records with a signup year, participation was higher in 2023 than in 2024.
Geography	The United States had the most participants, followed by India.
Institution	Saint Louis University had the most participants by a large margin.
Age and gender	Most learners were aged 20 to 30; male participation was higher than female participation.

Opportunity Category Results

Category	In Progress	Successful	Unsuccessful	Total
Competition	420	0	5	425
Course	1996	17	24	2037
Engagement	130	0	0	130
Event	531	12	2	545
Internship	1180	0	4235	5415

The clearest area for stakeholder attention is internship participation. Internships generated strong interest, but unsuccessful outcomes were much more common than in other categories. Courses, by comparison, combined high participation with a stronger outcome profile.

Predictive Modelling Approach

The model used opportunity, learner and timing information as input features. These included opportunity name and category, gender, country, intended major, age, application year and month, and signup year and month. The target was the learner's Status Group. Categorical fields

were converted into numeric indicators, and the data was divided into 80% training data and 20% testing data using a consistent random split.

Model Comparison

Model	Accuracy	Macro F1	Successful Recall	Overall Assessment
Logistic Regression	87.43%	64%	17%	Best overall balance
Decision Tree	88.25%	59%	0%	Highest accuracy, but missed all Successful cases
Random Forest	83.99%	60%	50%	Found more Successful cases, but with many false positives

Logistic Regression was selected as the final model. Although the Decision Tree achieved slightly higher accuracy, it did not correctly identify any Successful records. Random Forest identified three of the six Successful records, but only 7% of its Successful predictions were correct. Logistic Regression offered the strongest overall balance and the highest macro-average F1-score.

Recommendations for Stakeholders

- Prioritize early support for learners predicted as Unsuccessful through reminders, guidance and timely follow-up.
- Review the internship pathway to understand why unsuccessful outcomes are much higher than in other opportunity categories.
- Use course participation practices as a reference when improving other opportunity categories.
- Collect more completed and Successful records before using the model for high-impact decisions.
- Use model predictions as a screening tool alongside staff review, not as an automatic final decision.
- Monitor model performance regularly as new learner records become available.

Important Limitations

The Successful group was extremely small. The test dataset contained only six Successful records, compared with 852 In Progress and 853 Unsuccessful records. This imbalance made it difficult for every model to learn the Successful category reliably. Some date fields were also missing for many records, and the model results reflect the quality and coverage of the available data.

Conclusion

The analysis shows that the RIT dataset can provide useful guidance for improving student participation. Internship outcomes require the most attention, while courses show a stronger participation pattern. Logistic Regression is the recommended model from the three models tested because it produced the most balanced performance. With more complete data and more Successful examples, the model could become a stronger tool for early learner support and resource planning.